

Assignment Questions

No. 1

Observe the structures of a given EMPLOYEE table as below and suggest if this table can be further Normalized

EMPNO, ENAME, SAL, DEPTNO, DNAME, LOC

Explain what normal form this table is and how you can make this into next normal form.

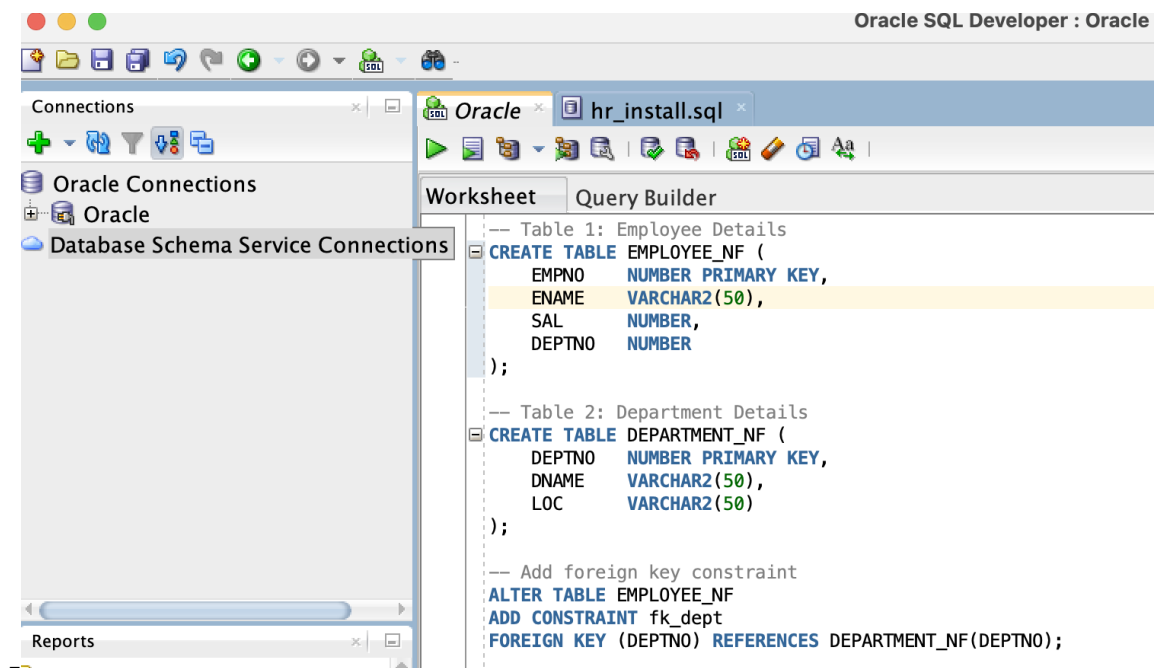
Solution :Assignment 1: EMPLOYEE Table (EMPNO, ENAME, SAL, DEPTNO, DNAME, LOC)

Current Normal Form: 2NF (Second Normal Form)

Reason:

- EMPNO is the primary key.
- ENAME, SAL depend on EMPNO (fully).
- But DNAME and LOC depend on DEPTNO, not on EMPNO → partial dependency.
- So it's in 1NF and 2NF (since no partial dependency on part of a composite key, but here PK is single), but **not in 3NF** because of transitive dependency: EMPNO → DEPTNO → DNAME, LOC.

Solution: Decompose into 2 tables to achieve 3NF.



Assignment 2

Observe the structure of a given STUDENT table as below and suggest if this table can be further Normalized

ROLLNO, NAME, AGE, EXAM, MARKS, GRADE

Explain what normal form this table is and how you can make this into next normal form

Solution: Assignment 2: STUDENT Table (ROLLNO, NAME, AGE, EXAM, MARKS, GRADE)

Current Normal Form: 1NF (First Normal Form)

Reason:

- ROLLNO alone cannot be primary key because one student can appear in multiple exams.
- Composite key: (ROLLNO, EXAM)
- NAME, AGE depend only on ROLLNO (partial dependency) → not in 2NF.

Solution: Decompose into 3 tables to achieve 3NF.

The screenshot shows the Oracle SQL Developer interface. The main window displays the SQL script for creating three tables: STUDENT_NF, EXAM_NF, and MARKS_NF. The script is as follows:

```
CREATE TABLE STUDENT_NF (  
  ROLLNO NUMBER(10),  
  NAME VARCHAR2(50),  
  AGE NUMBER(3),  
  CONSTRAINT pk_student PRIMARY KEY (ROLLNO),  
);  
  
-- Table 2: Exam Master  
CREATE TABLE EXAM_NF (  
  EXAM VARCHAR2(50) PRIMARY KEY,  
);  
  
-- Table 3: Marks (fact table)  
CREATE TABLE MARKS_NF (  
  ROLLNO NUMBER,  
  EXAM VARCHAR2(50),  
  MARKS NUMBER,  
  GRADE CHAR(2),  
  CONSTRAINT pk_marks PRIMARY KEY (ROLLNO, EXAM),  
  CONSTRAINT fk_student FOREIGN KEY (ROLLNO) REFERENCES STUDENT_NF(ROLLNO),  
  CONSTRAINT fk_exam FOREIGN KEY (EXAM) REFERENCES EXAM_NF(EXAM)  
);
```

The Script Output window at the bottom shows the execution results:

```
Table STUDENT_NF created.  
Table EXAM_NF created.  
Table MARKS_NF created.
```

Assignment 3

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Observe the structure of a given EMPLOYEE table and suggest what kind of problem this table has and how to solve the problem
EMPNO,PROJECT_NO,NO_OF_DAYS,CUSTOMERNAME
In the above table EMPNO AND PROJECT_NO both are together a composite primary key

Solution: Assignment 3: EMPLOYEE Table (EMPNO, PROJECT_NO, NO_OF_DAYS, CUSTOMERNAME)

Primary Key: Composite (EMPNO, PROJECT_NO)

Problem: CUSTOMERNAME depends only on PROJECT_NO, not on the full composite key → **Partial dependency** → Table is in **1NF**, not in **2NF**.

Issues:

- Data redundancy (customer name repeated for each employee on same project).
- Update anomalies (changing customer name requires updating multiple rows).

Solution: Decompose into 2 tables to achieve **2NF** and **3NF**.

